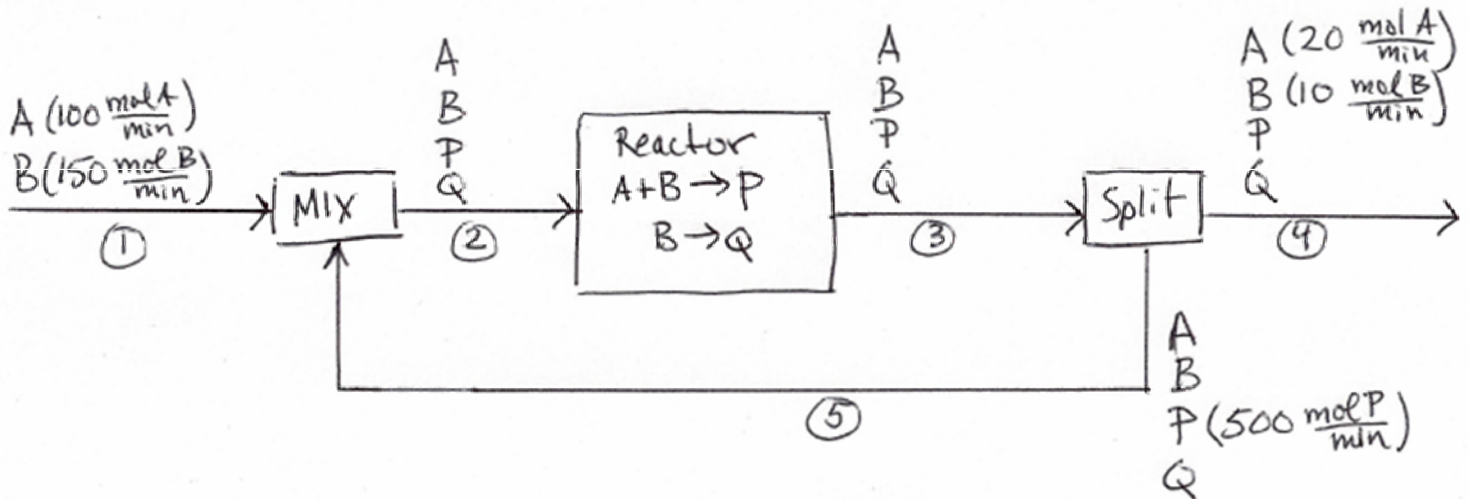


Consider the block flow diagram below. Notice that the streams are labeled 1, 2, 3, 4, and 5. Species molar flow rates are given by the species symbol with the stream number as a subscript. The fresh feed contains species A and B with species molar flow rates $A_1 = 100 \text{ molA/min}$ and $B_1 = 150 \text{ molB/min}$. This stream is mixed with a recycle stream containing A, B, P, and Q with the species molar flow rate of $P_5 = 500 \text{ molP/min}$. The mixture enters a reactor where two reactions take place: $A + B \rightarrow P$ and $B \rightarrow Q$. Species P is the desired product and species Q is a byproduct. The outlet stream from the reactor goes to a splitter with a product stream containing all four species with the species flow rates of $A_4 = 20 \text{ molA/min}$ and $B_4 = 10 \text{ molB/min}$.

- (a) Find all of the species molar flow rates not given in the problem statement. Fill in the table below. (8 pt.)
 (b) Calculate the overall and single pass fractional conversions based on species A. Calculate the fractional selectivity based on species B using the definition on page 299 of Murphy (2 pts.)



To receive full credit, fill in this table with the appropriate species flow rates in mole/min. Either submit an image of this table completed, or reproduce the table on your own paper and fill it in. Be sure to include the completed table with your PDF submission.

↓Species/Stream # →	1	2	3	4	5
A	100			20	
B	150			10	
P	----				500
Q	----				